# React Components

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Components are like functions that return HTML elements.

## React Components

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Components are independent and reusable bits of code. They serve the same purpose as JavaScript functions, but work in isolation and return HTML.

Components come in two types, Class components and Function components, in this tutorial we will concentrate on Function components.

In older React code bases, you may find Class components primarily used.

It is now suggested to use Function components along with Hooks, instead of Class components.

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# Create Your First Component

When creating a React component, the component's name *MUST* start with an upper case letter.

React components returns HTML code.

## Example

Get your own React.js Server

Create a Function component called `Car`

```
function Car() {
  return (
    <h2>Hi, I am a Car!</h2>
  );
}
```

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# Rendering a Component

Now your React application has a component called `Car`, which returns an `<h2>` element.

To use this component in your application, refer to it like this: `<Car />`

## Example

Display the `Car` component in the "root" element:

```
createRoot(document.getElementById('root')).render(
  <Car />
)
```

Run Example »

# Props

Arguments can be passed into a component as `props`, which stands for properties.

You send the arguments into the component as HTML attributes.

You will learn more about `props` in our React Props chapter.

## Example

Use an attribute to pass a color to the `Car` component, and use it in the `render` function:

```
function Car(props) {
  return (
```

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```
createRoot(document.getElementById('root')).render(
  <Car color="red"/>
);
```

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# Components in Components

We can refer to components inside other components:

## Example

Use the `Car` component inside the `Garage` component:

```
function Car() {
  return (
```

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```
function Garage() {
  return (
    <>
      <h1>Who lives in my Garage?</h1>
      <Car />
    </>
  );
}

createRoot(document.getElementById('root')).render(
  <Garage />
);
```

Run Example »

# Rendering a Component Twice

We can render a component multiple times:

## Example

Use the `Car` component twice inside the `Garage` component:

```
function Car() {
  return (
    <h2>I am a Car!</h2>
  );
}

function Garage() {
  return (
    <>
```

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```
  );
}

createRoot(document.getElementById('root')).render(
  <Garage />
);
```

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The example above might be a bit useless, but if we change the content of the `Car` component, by using arguments, it makes more sense:

## Example

Use the `Car` component to display two different cars:

```
function Car(props) {
  return (
    <h2>I am a {props.brand}!</h2>
  );
}

function Garage() {
  return (
    <>
      <h1>Who lives in my Garage?</h1>
      <Car brand="Ford" />
      <Car brand="BMW" />
    </>
  );
}

createRoot(document.getElementById('root')).render(
```

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# Components in Files

React is all about re-using code, and it can be a good idea to split your components into separate files.

To do that, create a new file in the `src` folder with a `.jsx` file extension and put the code inside it:

Note that the filename must start with an uppercase character.

## Example

This is the new file, we named it `Vehicle.jsx`:

Vehicle.jsx

```
function Car() {
  return (
    <h2>Hi, I am a Car!</h2>
  );
}

export default Car;
```

To be able to use the `Car` component, you have to import the `Vehicle.jsx` file in your application.

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as if it was created here.

main.jsx

```
import { createRoot } from 'react-dom/client'
import Car from './Vehicle.jsx';

createRoot(document.getElementById('root')).render(
  <Car />
);
```

Run Example »

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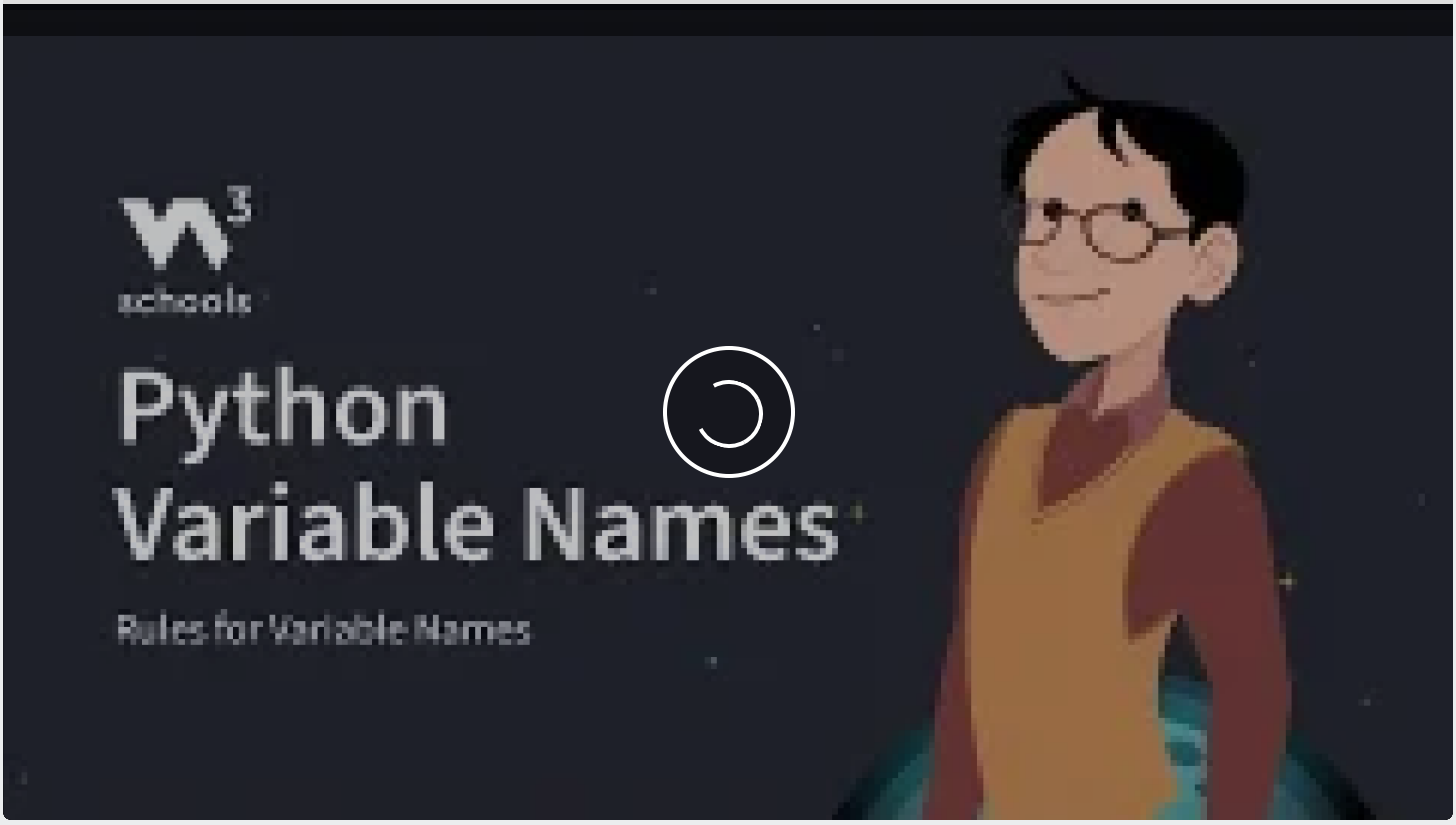
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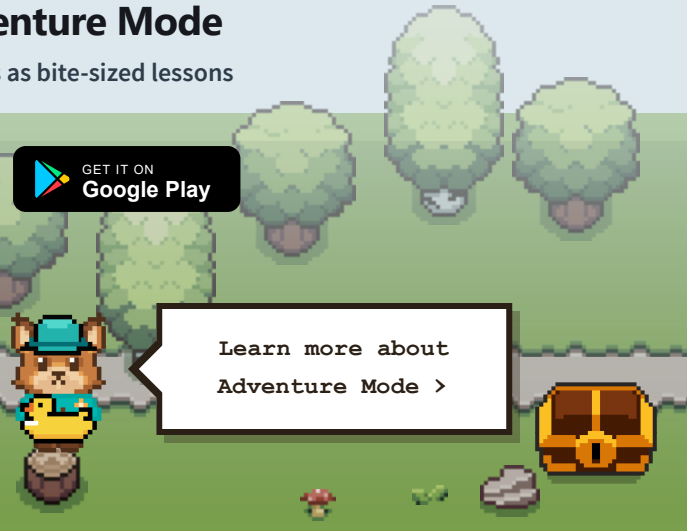
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